2. In a bungee jump, a light elastic rope is used to attach a man to a bridge. The man has a mass of 80.0 kg while the rope has a natural length of 25.0 m and an elastic constant of 120 N m^{-1}. The man steps off the bridge and falls vertically downwards from rest. Assume that air resistance acting on the man is negligible.

(a) **(i)** Show that the extension of the rope is 6.54 m when the man is falling at maximum speed.

[1]

(ii) Determine the maximum speed of the man after he steps off the bridge.

maximum speed = ... m s^{-1} [2]

(iii) Calculate the maximum kinetic energy of the man.

maximum kinetic energy = ... J [1]

(b) **(i)** Calculate the maximum extension of the elastic rope when the man is at the lowest point of his motion.

maximum extension = ... m [2]

(ii) Hence, or otherwise, determine the gravitational potential energy of the man on the bridge. Assume that gravitational potential energy of the man is zero at the lowest point of the man's motion.

gravitational potential energy = ... J [1]

(c) On Fig. 2, sketch three clearly-labelled graphs for the variation with downward displacement *s* of

(i) the gravitational potential energy of the man, (Label as **G**)

(ii) the elastic potential energy stored in the rope, and (Label as **E**)

(iii) the kinetic energy of the man. (Label as **K**)

Assume that gravitational potential energy of the man is zero at the lowest point of the man's motion. Take s = 0 m as the start point of motion.

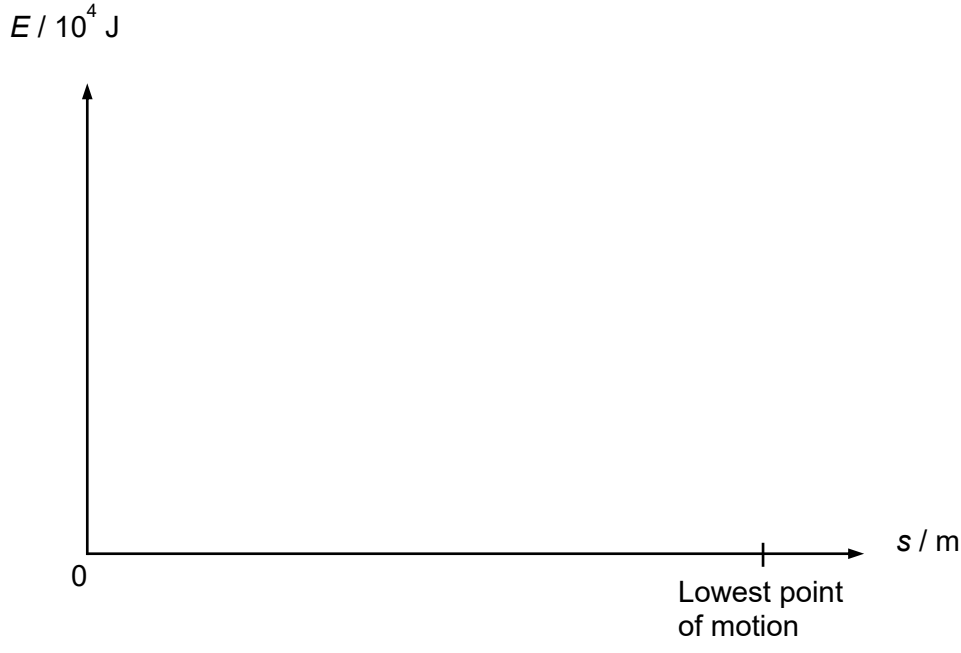

Fig. 2

[3]